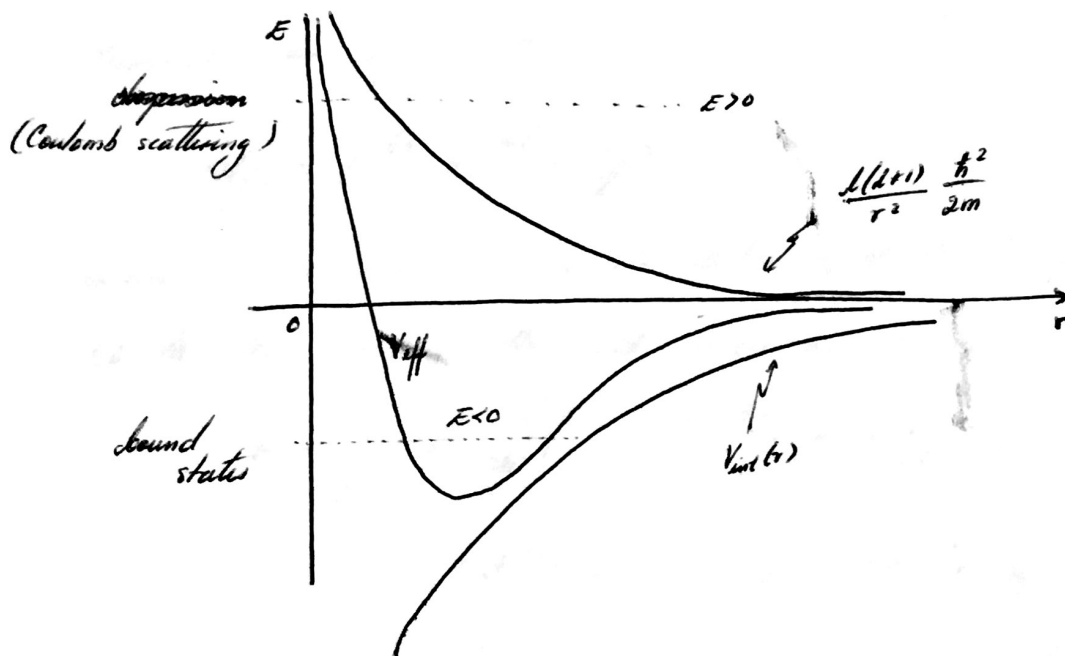


~ The hydrogen potential
The hydrogen atom is governed by the ^{electrostatic} potential

$$V_{int}(r) = -\frac{e^2}{4\pi\epsilon_0 r}$$

$$\Rightarrow V_{eff} = -\frac{e^2}{4\pi\epsilon_0 r} + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2}$$



We're interested only in bound states of the hydrogen atom, i.e. $E < 0$.

It's convenient to propose adimensional coordinates

$$\rho \equiv 2\alpha r \quad \alpha \equiv \sqrt{\frac{2m|E|}{\hbar^2}} \quad \text{or} \quad \alpha^2 \equiv -\frac{2mE}{\hbar^2}$$

$$\Rightarrow \frac{d^2 u}{dr^2} = \left(\frac{d\rho}{dr}\right)^2 \frac{d^2 u}{d\rho^2} = 4\alpha^2 u''$$

$$\Rightarrow 4\alpha^2 u'' + \left[\frac{2m}{\hbar^2} \left(\underbrace{E}_{=-|E|} + \frac{e^2 2\alpha}{4\pi\epsilon_0 \rho} \right) - 4\alpha^2 \frac{l(l+1)}{\rho^2} \right] u = 0$$

$$\Rightarrow 4\alpha^2 \left\{ u'' + \left[\frac{-1}{4} + \frac{1}{2\alpha} \frac{e^2}{4\pi\epsilon_0 \rho} \frac{2m}{\hbar^2} - \frac{l(l+1)}{\rho^2} \right] u \right\} = 0$$

$$\Leftrightarrow u'' + \left[\frac{-1}{4} + \frac{me^2}{4\pi\epsilon_0 \hbar^2} \frac{1}{\alpha \rho} - \frac{l(l+1)}{\rho^2} \right] u = 0 \quad a_0: \text{Bohr radius} \quad a_0^{-1} \equiv \frac{1}{4\pi^2} \frac{me^2 \pi}{\epsilon_0 \hbar^2} = \frac{me^2 \pi}{\epsilon_0 \hbar^2}$$

We have used that $m \approx m_e$.

(71)

To solve this equation, it's useful to observe that

$$\frac{d^2 u}{dr^2} - \frac{l(l+1)}{r^2} u = 0 \quad \Rightarrow \quad u \sim r^{l+1}$$

$$\& \quad -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} = E u \quad \Rightarrow \quad u \sim e^{-\alpha r} \quad \alpha^2 = -\frac{2mE}{\hbar^2}$$

Therefore, we expect that

$$u(\rho) = \rho^{l+1} e^{-\frac{1}{2}\rho} Q(\rho)$$

where $Q(\rho)$ must capture the behavior dictated by the $\frac{1}{\rho}$ term.

With this ansatz, the Schrödinger eq. becomes

$$\begin{aligned} u'' &= \frac{d^2}{d\rho^2} u(\rho) = \frac{d}{d\rho} \left((l+1)\rho^l e^{-\frac{1}{2}\rho} Q(\rho) - \frac{1}{2}\rho^{l+1} e^{-\frac{1}{2}\rho} Q(\rho) + \rho^{l+1} e^{-\frac{1}{2}\rho} Q'(\rho) \right) \\ &= l(l+1)\rho^{l-1} e^{-\frac{1}{2}\rho} Q(\rho) - \frac{1}{2}(l+1)\rho^l e^{-\frac{1}{2}\rho} Q + (l+1)\rho^l e^{-\frac{1}{2}\rho} Q' \\ &\quad - \frac{1}{2}(l+1)\rho^l e^{-\frac{1}{2}\rho} Q + \frac{1}{2}\rho^{l+1} e^{-\frac{1}{2}\rho} Q - \frac{1}{2}\rho^{l+1} e^{-\frac{1}{2}\rho} Q' + (l+1)\rho^l e^{-\frac{1}{2}\rho} Q' \\ &\quad - \frac{1}{2}\rho^{l+1} e^{-\frac{1}{2}\rho} Q' + \rho^{l+1} e^{-\frac{1}{2}\rho} Q'' \\ &= \frac{l(l+1)}{\rho^2} u + \frac{1}{4} u - \frac{(l+1)u}{\rho} + \frac{2(l+1)}{\rho} \frac{u}{Q} Q' - u \frac{Q'}{Q} + u \frac{Q''}{Q} \end{aligned}$$

$$\Rightarrow u \left\{ \frac{Q''}{Q} + \left(\frac{1}{\rho^2} - (l+1) \right) \frac{1}{\rho} + \left(\frac{2(l+1)}{\rho} - 1 \right) \frac{Q'}{Q} \right\} = 0$$

$$\Leftrightarrow Q'' + \left(\frac{2(l+1)}{\rho} - 1 \right) Q' + \frac{(l_0 \alpha)^{-1} - l - 1}{\rho} Q = 0$$

$$\Leftrightarrow \rho Q'' + (2(l+1) - \rho) Q' + (l_0 \alpha)^{-1} - l - 1) Q = 0$$

which clearly resembles the Laguerre equation

$$\rho L_k''(\rho) + (s+1-\rho)L' + kL = 0$$

whose solutions only converge for

$$k, n \in \mathbb{Z} \quad k \geq 0$$

and one called the associated Laguerre polynomials given by the Rodrigues formula

$$L_k^s(\rho) = \frac{e^\rho \rho^{-s}}{k!} \frac{d^k}{d\rho^k} (e^{-\rho} \rho^{k+s})$$

for our case, $k \Rightarrow (a_0 \alpha)^{-1} - l - 1$ & $s \Rightarrow 2l+1$ $l \geq 0$

$$\Rightarrow k = (a_0 \alpha)^{-1} - l - 1 \geq 0 \Rightarrow \begin{cases} (a_0 \alpha)^{-1} = \overbrace{k+l+1}^{\text{these are } \mathbb{N}} \geq 1 & \Rightarrow (a_0 \alpha)^{-1} \equiv n \in \mathbb{N}^+ \\ \boxed{l \leq n-1} \end{cases} \rightarrow 1, 2, 3, \dots$$

n : ~~main~~ principal quantum number

$$n = k+l+1 \geq 1 \rightarrow 1, 2, \dots$$

$$\Rightarrow l = n-k-1$$

$$\Rightarrow l \leq n-1$$

$$Q = L_k^{2l+1} = \frac{e^\rho \rho^{-2l-1}}{k!} \frac{d^k}{d\rho^k} (e^{-\rho} \rho^{k+2l+1})$$

Notice that

$$n = \frac{1}{a_0 \alpha} = \frac{me^2}{4\pi\epsilon_0 \hbar^2} \sqrt{\frac{\hbar^2}{2m|E|}} = \sqrt{\frac{-me^4}{8\pi^2\epsilon_0^2 \hbar^2 E}} \Rightarrow E_n = -\frac{\hbar^2}{2m a_0^2} \frac{1}{n^2} = -\frac{e^2}{8\pi\epsilon_0 a_0} \frac{1}{n^2}$$

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{me^2}$$

see page (15)
coincides!

$$\Rightarrow \boxed{E_n = -\frac{1}{4\pi^2} \frac{me^4}{8\epsilon_0^2 \hbar^2} \frac{1}{n^2}} = -\frac{me^4}{8\epsilon_0^2 \hbar^2} \frac{1}{n^2}$$

(The factor $\hbar^2$ is ~~cancelled~~ due to $\hbar = \frac{h}{2\pi}$) (see also def. of a_0)

Thus, using $R = r/r$

$$R_{nl}(\rho) = C_{nl} \rho^l e^{-\frac{1}{2}\rho} L_{n-l-1}^{2l+1}(\rho), \quad l = 0, 1, 2, \dots, n-1$$

$$\Rightarrow \psi_{nl}^m(\rho, \vartheta, \varphi) = C_{nl} \rho^l e^{-\frac{1}{2}\rho} L_{n-l-1}^{2l+1}(\rho) Y_l^m(\vartheta, \varphi)$$

subject to the condition

$$\int r^2 dr \sin\vartheta d\vartheta d\varphi |\psi_{nl}^m|^2 = 1$$

$$\Rightarrow \frac{1}{8a^3} \int_0^\infty \rho^2 d\rho |C_{nl}|^2 \rho^{2l} e^{-\rho} \left(L_{n-l-1}^{2l+1}(\rho) \right)^2 \underbrace{\int \sin\theta d\theta d\varphi |Y_l^m(\theta, \varphi)|^2}_{=1} = 1$$

The orthogonality condition of ^{ass.} Laguerre polynomials read:

$$\int_0^\infty d\rho \rho^s e^{-\rho} L_k^s(\rho) L_{k'}^s(\rho) = \frac{\Gamma(k+s+1)}{k!} \delta_{k,k'}$$

where for $k+s+1$ integer & positive ~~and~~ the gamma function is

$$\Gamma(k+s+1) = (k+s)!$$

(For other values of $t=k+s+1$, it can be written as $\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$)

Using recurrence relations, one can also show that

$$\int_0^\infty d\rho e^{-\rho} \rho^{s+1} \left(L_k^s(\rho) \right)^2 = \frac{(k+s)!}{k!} (2k+s+1)$$

$$\Rightarrow \frac{1}{8a^3} |C_{nl}|^2 \int d\rho \rho^{(2l+1)+1} e^{-\rho} \left(L_{n-l-1}^{2l+1}(\rho) \right)^2 = 1$$

$$\Leftrightarrow \frac{|C_{nl}|^2}{8a^3} \frac{(n+l)!}{(n-l-1)!} (2n) = 1$$

$$\frac{1}{a_0 n} = \alpha \quad \Leftrightarrow \frac{|C_{nl}|^2}{4} (a_0 n)^3 n \frac{(n+l)!}{(n-l-1)!} = 1$$

$$\Leftrightarrow C_{nl} = \frac{2}{n^2} \sqrt{\frac{(n-l-1)!}{a_0^3 (n+l)!}} \quad \text{up to an unphysical phase}$$

$$\Rightarrow \boxed{\psi_{nl}^m(\rho, \theta, \varphi) = \frac{2}{n^2} \sqrt{\frac{(n-l-1)!}{a_0^3 (n+l)!}} \rho^l e^{-\frac{1}{2}\rho} L_{n-l-1}^{2l+1}(\rho) Y_l^m(\theta, \varphi)}$$

Recommended literature:

- de la Peña, § 13 (Hydrogen atom); § App. A (Polynomials)
- Cohen-Tannoudji, § VII A (Central potential); § VII C (Hydrogen atom)
- Beiser, § 6 (Intro. to hydrogen atom)

Each of these normalized wavefunctions corresponds to a state of the "mass-reduced" electron in the hydrogen atom. However, since the energy E_n depends only on the principal quantum number n , there is a multiplicity of (counting all possible values of l and m for each n) degeneracy

$$\sum_{l=0}^{n-1} (2l+1) = n^2$$

states with the same energy. This is called the degeneracy of the energy level.

This degeneracy is associated with two quantizations:

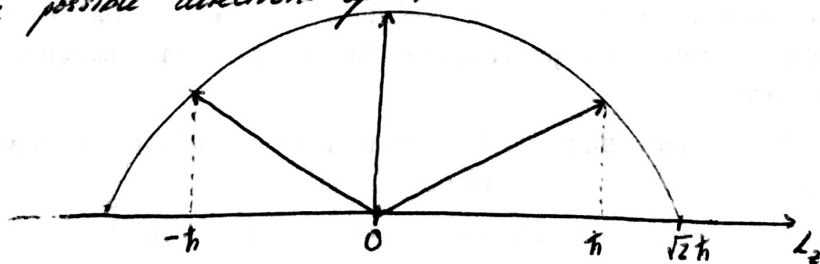
l (orbital quantum number)

Since $\hat{L}^2 \psi = \hbar^2 l(l+1) \psi$, the values that l can take show that the magnitude of $\vec{L}$ is quantized, such that $|\vec{L}| = \sqrt{l(l+1)} \hbar$
(similar to Bohr's condition)
(for $l_{max} = n-1$ $|\vec{L}| = n\hbar \sqrt{1 - \frac{1}{n^2}}$)

m (magnetic quantum number)

Since $\hat{L}_z \psi = \hbar m \psi$, the dif. values that m takes refer to the few directions that the $\vec{L}$ can take, provided n and l ; i.e. the direction of $\vec{L}$ is quantized.

Take e.g. $l=1 \Rightarrow |\vec{L}| = \sqrt{1(1+1)} \hbar = \sqrt{2} \hbar$
 $\rightarrow$ the possible directions of $\vec{L}$ are



$$L_z = 0, \pm \hbar$$

Notation: it's customary to use the historical notation for the states $n(\text{letter})$

where

l	0	1	2	3	4	...
letter	s	p	d	f	g	

E.g. states

2p, 3s, 4f are possible

1s corresponds to the ground state

To conclude, note that for $l_{\max} = n-1$

$$\rho R(\rho) \propto \rho^{l_{\max}+1} e^{-\frac{1}{2}\rho}$$

which has a maximum ~~at~~ when

$$\left[(l_{\max}+1) \rho^{l_{\max}} - \frac{1}{2} \rho^{l_{\max}+1} \right] e^{-\frac{1}{2}\rho} = 0$$

$$\Leftrightarrow n - \frac{1}{2} 2 \alpha r_{\max} = 0 \quad \Leftrightarrow n = \frac{1}{na_0} r_{\max} \quad \Leftrightarrow \boxed{r_{\max} = n^2 a_0}$$

coinciding with Bohr's estimate of H-atom's radius.

3.3 ~ Selection rules & spectral series

The state associated with the wave function

$$\psi_{nl}^m = R_{nl} Y_l^m$$

can be expressed as

$$|nlm\rangle = |nl\rangle |lm\rangle$$

such that

$$\langle r, \vartheta, \varphi | nlm \rangle = \langle r | nl \rangle \langle \vartheta, \varphi | lm \rangle = R_{nl}(r) Y_l^m(\vartheta, \varphi)$$

Our goal now is to determine which transitions

$$|nlm\rangle \rightarrow |n'l'm'\rangle$$

may occur by emitting some dipole radiation, i.e. we have to determine the values of n, l, m for which

$$\mathcal{M}_{nlm, n'l'm'} = \frac{4e^2 \omega^3}{3\hbar c^3} |\langle n'l'm' | \hat{r} | nlm \rangle|^2 \neq 0$$

is satisfied. Recall that

$$\vec{r} = r (\sin\vartheta \cos\varphi \vec{e}_1 + \sin\vartheta \sin\varphi \vec{e}_2 + \cos\vartheta \vec{e}_3)$$

$$\Rightarrow \langle n'l'm' | \hat{r} | nlm \rangle = \langle n'l' | \hat{r} | n l \rangle \langle l'm' | \sin\theta \cos\varphi \vec{e}_x + \sin\theta \sin\varphi \vec{e}_y + \cos\theta \vec{e}_z | l m \rangle$$

$$= \left[\int_0^\infty r^2 dr R(r)_{n'l'} R_{nl}(r) r \right] \langle l'm' | \dots | l m \rangle$$

$\neq 0 \quad \forall n, n'$

Thus, only the condition

$$\langle r | n l \rangle = R_{nl}(r)$$

$$\langle \vartheta, \varphi | l m \rangle = Y_l^m(\vartheta, \varphi)$$

$$\langle l'm' | \sin\theta \cos\varphi \vec{e}_x + \sin\theta \sin\varphi \vec{e}_y + \cos\theta \vec{e}_z | l m \rangle \neq 0$$

maths. It's convenient to rewrite the operator as

$$\langle l'm' | \sin\theta e^{i\varphi} \vec{e}_+ + \sin\theta e^{-i\varphi} \vec{e}_- + \cos\theta \vec{e}_z | l m \rangle \neq 0$$

$$Y_l^{m*} = (-1)^m Y_l^{-m}$$

with $\vec{e}_\pm \equiv \frac{1}{2}(\vec{e}_x \mp i\vec{e}_y)$ describing rotations clockwise (anticlockwise) rotations around the $\hat{z}$ -axis.

$$\Rightarrow \langle l'm' | \dots | l m \rangle = \int d\Omega Y_{l'}^{m'*} \sin\theta e^{i\varphi} Y_l^m \vec{e}_+ + \int d\Omega Y_{l'}^{m'*} \sin\theta e^{-i\varphi} Y_l^m \vec{e}_- + \int d\Omega Y_{l'}^{m'*} \cos\theta Y_l^m \vec{e}_z$$

Term by term: (use integrals of 3 Y 's in terms of Wigner 3j-symbol)*

$$\textcircled{1} \vec{e}_+ \int_0^\pi \sin^2\theta d\theta \int_0^{2\pi} d\varphi e^{i\varphi} Y_{l'}^{m'*} Y_l^m = \vec{e}_+ \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\varphi Y_{l'}^{m'*} \left(-2\sqrt{\frac{2\pi}{3}} Y_1^1 \right) Y_l^m$$

$$Y_1^1 = -\sqrt{\frac{3}{8\pi}} \sin\theta e^{i\varphi}$$

$$= -2\sqrt{\frac{2\pi}{3}} \vec{e}_+ \sqrt{\frac{3}{8\pi}} \left[\sqrt{\frac{(l+m+1)(l+m+2)}{(2l+1)(2l+3)}} \delta_{l', l+1} - \sqrt{\frac{(l-m)(l-m-1)}{(2l+1)(2l+3)}} \delta_{l', l-1} \right] \delta_{m', m+1}$$

$$m = -1 + m' \Rightarrow m' = -1 - m$$

$$\textcircled{2} \vec{e}_- \int d\Omega \sin\theta e^{-i\varphi} Y_{l'}^{m'*} Y_l^m = \vec{e}_- \int d\Omega Y_{l'}^{m'*} \left(2\sqrt{\frac{2\pi}{3}} Y_1^{-1} \right) Y_l^m$$

$$Y_1^{-1} = \sqrt{\frac{3}{8\pi}} \sin\theta e^{-i\varphi}$$

$$= 2\sqrt{\frac{2\pi}{3}} \vec{e}_- \sqrt{\frac{3}{8\pi}} \left[\sqrt{\frac{(l-m+2)(l-m+1)}{(2l+1)(2l+3)}} \delta_{l', l+1} - \dots \right]$$

* Use $\int Y_{l_1}^{m_1} Y_{l_2}^{m_2} Y_{l_3}^{m_3} = \sqrt{\frac{(2l_1+1)(2l_2+1)(2l_3+1)}{4\pi}} \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix}$, $l_3 = l_1 + l_2$
 $m_3 = -m_1 - m_2$

with $\begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{l_1 - l_2 + m_3} \frac{(2l_1)!(2l_2)!}{(2l_1 + 2l_2 + 1)!} \frac{(l_1 + l_2 + m_3)!(l_1 + l_2 + m_3)!}{(l_1 + m_1)!(l_1 - m_1)!(l_2 + m_2)!(l_2 - m_2)!}$

$$\Rightarrow \langle n'l'm' | \hat{r} | nlm \rangle = \langle n'l' | \hat{r} | nl \rangle \langle l'm' | \sin\theta \cos\varphi \hat{e}_x + \sin\theta \sin\varphi \hat{e}_y + \cos\theta \hat{e}_z | lm \rangle$$

$$= \left[\int_0^\infty r^2 dr R(r)_{n'l'} R_{nl}(r) r \right] \langle l'm' | \dots | lm \rangle$$

$\neq 0 \quad \forall n, n'$

$$\langle r | nl \rangle = R_{nl}(r)$$

Thus, only the condition

$$\langle \theta, \varphi | lm \rangle = Y_l^m(\theta, \varphi)$$

$$\langle l'm' | \sin\theta \cos\varphi \hat{e}_x + \sin\theta \sin\varphi \hat{e}_y + \cos\theta \hat{e}_z | lm \rangle \neq 0$$

holds. It's convenient to rewrite the operator as

$$\langle l'm' | \sin\theta e^{i\varphi} \hat{e}_+ + \sin\theta e^{-i\varphi} \hat{e}_- + \cos\theta \hat{e}_z | lm \rangle \neq 0$$

$$Y_l^{m*} = (-1)^m Y_l^{-m}$$

with $\hat{e}_\pm = \frac{1}{2}(\hat{e}_x \mp i\hat{e}_y)$ describing rotations clockwise (anticlockwise) rotations around the $\hat{e}_z$ -axis.

$$\Rightarrow \langle l'm' | \dots | lm \rangle = \int d\Omega Y_l^{m'*} \sin\theta e^{i\varphi} Y_l^m \hat{e}_+ + \int d\Omega Y_l^{m'*} \sin\theta e^{-i\varphi} Y_l^m \hat{e}_- + \int d\Omega Y_l^{m'*} \cos\theta Y_l^m \hat{e}_z$$

Term by term: (use integrals of 3 Y 's in terms of Wigner 3j-symbols)*

$$\textcircled{1} \hat{e}_+ \int \sin^2\theta d\theta \int d\varphi e^{i\varphi} Y_l^{m'*} Y_l^m = \hat{e}_+ \int_0^{2\pi} d\varphi \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\varphi Y_l^{m'*} \left(-2\sqrt{\frac{2}{3}} Y_1^{-1} \right) Y_l^m$$

$$Y_1^{-1} = -\sqrt{\frac{3}{8\pi}} \sin\theta e^{i\varphi}$$

$$= -2\sqrt{\frac{2}{3}} \hat{e}_+ \sqrt{\frac{3}{8\pi}} \left[\sqrt{\frac{(l+m+1)(l+m+2)}{(2l+1)(2l+3)}} \delta_{l',l+1} - \sqrt{\frac{(l-m)(l-m-1)}{(2l+1)(2l+3)}} \delta_{l',l-1} \right] \delta_{m',m+1}$$

$$m = -1 + m'$$

$$m' = -1 - m$$

$$\textcircled{2} \hat{e}_- \int d\Omega \sin\theta e^{-i\varphi} Y_l^{m'*} Y_l^m = \hat{e}_- \int d\Omega Y_l^{m'*} \left(2\sqrt{\frac{2}{3}} Y_1^{-1} \right) Y_l^m$$

$$Y_1^{-1} = \sqrt{\frac{3}{8\pi}} \sin\theta e^{-i\varphi}$$

$$= 2\sqrt{\frac{2}{3}} \hat{e}_- \sqrt{\frac{3}{8\pi}} \left[\sqrt{\frac{(l-m+2)(l-m+1)}{(2l+1)(2l+3)}} \delta_{l',l+1} - \sqrt{\frac{(l+m)(l+m+1)}{(2l+1)(2l+3)}} \delta_{l',l-1} \right] \delta_{m',m-1}$$

* the $\int Y_{l_1}^{m_1} Y_{l_2}^{m_2} Y_{l_3}^{m_3} = \sqrt{\frac{(2l_1+1)(2l_2+1)(2l_3+1)}{4\pi}} \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix}$, $l_3 = l_1 + l_2$
 $m_3 = -m_1 - m_2$

with $\begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{l_1-l_2+m_3} \sqrt{\frac{(2l_1)!(2l_2)!}{(2l_1+2l_2+1)!}} \frac{(l_1+l_2+m_3)!(l_1+l_2+m_3)!}{(l_1+m_1)!(l_1-m_1)!(l_2+m_2)!(l_2-m_2)!}$

$$\Rightarrow \langle n'l'm' | \hat{r} | nlm \rangle = \langle n'l' | \hat{r} | nl \rangle \langle l'm' | \sin\delta \cos\varphi \vec{e}_x + \sin\delta \sin\varphi \vec{e}_y + \cos\delta \vec{e}_z | lm \rangle$$

$$= \left[\int_0^\infty r^2 dr R_{n'l'}^*(r) R_{nl}(r) r \right] \langle l'm' | \dots | lm \rangle$$

$\neq 0 \quad \forall n, n'$

Thus, only the condition

$$\langle r | nl \rangle = R_{nl}(r)$$

$$\langle \vartheta, \varphi | lm \rangle = Y_l^m(\vartheta, \varphi)$$

$$\langle l'm' | \sin\delta \cos\varphi \vec{e}_x + \sin\delta \sin\varphi \vec{e}_y + \cos\delta \vec{e}_z | lm \rangle \neq 0$$

matters. It's convenient to rewrite the operator as

$$\langle l'm' | \sin\delta e^{i\varphi} \vec{e}_+ + \sin\delta e^{-i\varphi} \vec{e}_- + \cos\delta \vec{e}_z | lm \rangle \neq 0$$

$$Y_l^{m*} = (-1)^m Y_l^{-m}$$

with $\vec{e}_\pm \equiv \frac{1}{2}(\vec{e}_x \mp i\vec{e}_y)$ describing rotations clockwise (anticlockwise) rotations around the $\vec{z}$ -axis.

$$\Rightarrow \langle l'm' | \dots | lm \rangle = \int d\Omega Y_{l'}^{m'*} \sin\delta e^{i\varphi} Y_l^m \vec{e}_+ + \int d\Omega Y_{l'}^{m'*} \sin\delta e^{-i\varphi} Y_l^m \vec{e}_-$$

$$+ \int d\Omega Y_{l'}^{m'*} \cos\delta Y_l^m \vec{e}_z$$

Term by term: (use integrals of 3 Y's in terms of Wigner 3j-symbol)*

$$\textcircled{1} \vec{e}_+ \int \sin^2\delta d\vartheta \int_0^{2\pi} d\varphi e^{i\varphi} Y_{l'}^{m'*} Y_l^m = \vec{e}_+ \int_0^\pi \sin\delta d\vartheta \int_0^{2\pi} d\varphi Y_{l'}^{m'*} \left(-2\sqrt{\frac{2\pi}{3}} Y_1^1 \right) Y_l^m$$

$$Y_1^1 = -\sqrt{\frac{3}{8\pi}} \sin\delta e^{i\varphi}$$

$$= -2\sqrt{\frac{2\pi}{3}} \vec{e}_+ \sqrt{\frac{3}{8\pi}} \left[\sqrt{\frac{(l+m+1)(l+m+2)}{(2l+1)(2l+3)}} \delta_{l', l+1} - \sqrt{\frac{(l-m)(l-m-1)}{(2l+1)(2l+3)}} \delta_{l', l-1} \right] \delta_{m', m+1}$$

$$\begin{aligned} m &= -l+m' \\ m' &= -l-m \end{aligned}$$

$$\textcircled{2} \vec{e}_- \int d\Omega \sin^2\delta e^{-i\varphi} Y_{l'}^{m'*} Y_l^m = \vec{e}_- \int d\Omega Y_{l'}^{m'*} \left(2\sqrt{\frac{2\pi}{3}} Y_1^{-1} \right) Y_l^m$$

$$Y_1^{-1} = \sqrt{\frac{3}{8\pi}} \sin\delta e^{-i\varphi}$$

$$= 2\sqrt{\frac{2\pi}{3}} \vec{e}_- \sqrt{\frac{3}{8\pi}} \left[\sqrt{\frac{(l-m+2)(l-m+1)}{(2l+1)(2l+3)}} \delta_{l', l+1} - \dots \right]$$

$$* \text{ Use } \int Y_{l_1}^{m_1} Y_{l_2}^{m_2} Y_{l_3}^{m_3} = \sqrt{\frac{(2l_1+1)(2l_2+1)(2l_3+1)}{4\pi}} \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix}, \quad \begin{aligned} l_3 &= l_1 + l_2 \\ m_3 &= m_1 + m_2 \end{aligned}$$

$$\text{with } \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{l_1-l_2+m_3} \sqrt{\frac{(2l_1)!(2l_2)!}{(2l_1+2l_2+1)!}} \frac{(l_1+l_2-m_3)!(l_1+l_2+m_3)!}{(l_1+m_1)!(l_1-m_1)!(l_2+m_2)!(l_2-m_2)!}$$

cont...

$$-\sqrt{\frac{(l+m)(l+m-1)}{(2l+1)(2l-1)}} \int \delta_{l',l-1} \delta_{m',m-1}$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos \vartheta$$

3

$$\vec{e}_2 \int d\Omega \cos \vartheta Y_{l'}^{m'} Y_l^m = \vec{e}_2 \int d\Omega Y_{l'}^{m'} \left(\sqrt{\frac{4\pi}{3}} Y_1^0 \right) Y_l^m$$

$$= \sqrt{\frac{4\pi}{3}} \vec{e}_2 \sqrt{\frac{3}{4\pi}} \left[\sqrt{\frac{(l+m+1)(l-m+1)}{(2l+3)(2l+1)}} \delta_{l',l+1} + \sqrt{\frac{(l+m)(l-m)}{(2l+1)(2l-1)}} \delta_{l',l-1} \right] \delta_{m',m}$$

$$\Rightarrow \langle l'm' | \dots | l m \rangle = \vec{e}_+ \left[\underbrace{\sqrt{\frac{(l-m)(l-m-1)}{(2l+1)(2l-1)}}}_{\equiv A_{m+1}^{l-1}} \delta_{l',l-1} - \underbrace{\sqrt{\frac{(l+m+1)(l+m+2)}{(2l+1)(2l+3)}}}_{\equiv A_{m+1}^{l+1}} \delta_{l',l+1} \right] \delta_{m',m+1}$$

$$+ \vec{e}_- \left[\underbrace{\sqrt{\frac{(l-m+2)(l-m+1)}{(2l+1)(2l+3)}}}_{\equiv A_{m+1}^{l-1}} \delta_{l',l+1} - \underbrace{\sqrt{\frac{(l+m)(l+m-1)}{(2l+1)(2l-1)}}}_{\equiv A_{m-1}^{l-1}} \delta_{l',l-1} \right] \delta_{m',m-1}$$

$$+ \vec{e}_z \left[\underbrace{\sqrt{\frac{(l+m+1)(l-m+1)}{(2l+3)(2l+1)}}}_{\equiv A_m^{l+1}} \delta_{l',l+1} + \underbrace{\sqrt{\frac{(l+m)(l-m)}{(2l+1)(2l-1)}}}_{\equiv A_m^{l-1}} \delta_{l',l-1} \right] \delta_{m',m}$$

Thus $\langle l'm' | \dots | l m \rangle \neq 0$ only for transitions in which

$$\Delta l = \pm 1 \quad \text{and} \quad \Delta m = 0, \pm 1 \quad \text{and} \quad \Delta n \in \mathbb{Z} \text{ arbitrary}$$

Spontaneous transitions are those such that $n' < n$ ($\Rightarrow E_{n'} < E_n$); the emitted photons have frequency

$$\omega_{nn'} = \frac{E_n - E_{n'}}{\hbar} = -\frac{1}{4\pi^2} \frac{me^4}{8\epsilon_0^2 \hbar^3} \left(\frac{1}{n^2} - \frac{1}{n'^2} \right)$$

$$= 2\pi c R \left(\frac{1}{n^2} - \frac{1}{n'^2} \right) \quad \begin{matrix} \text{speed of light} \\ \text{Rydberg constant} \end{matrix} \quad R \equiv \frac{me^4}{8\epsilon_0^2 \hbar^3 c} \quad \hbar = 2\pi \hbar$$

We recover here the well-known spectral series

- Lyman $n' = 1$
- Balmer $n' = 2$
- Paschen $n' = 3$

In the "new" notation for the states in terms of angular momentum quantum number l , in the Lyman series, the spectral lines come from transitions

$$2p \rightarrow 1s + \gamma$$

$$3p \rightarrow 1s + \gamma$$

$$4p \rightarrow 1s + \gamma$$

$$\text{Since } \Delta l = 1 \text{ \& } \frac{dL}{dt} = 0 \Rightarrow l_y = 1$$

Note that transitions such as

$$3s \rightarrow 1s \quad \text{or} \quad 4d \rightarrow 1s \quad \times$$

are not allowed due to the selection rules. Also observe the degeneracy due to m .

In the Balmer series, there are more options:

$$3p, 4p, 5p, \dots \rightarrow 2s \quad \Delta l = 1$$

$$3s, 4s, 5s, \dots \rightarrow 2p \quad \Delta l = -1$$

$$3d, 4d, 5d, \dots \rightarrow 2p \quad \Delta l = 1$$

We can compute the ~~prob~~ transition amplitude of ~~any~~ transitions with $\Delta l = l_{\text{fin}} - l_{\text{ini}} = +1$ (for a fixed n)

$$|\langle l+1, m' | \dots | l, m \rangle|^2 = |\langle l+1, m+1 | \dots | l, m \rangle + \langle l+1, m-1 | \dots | l, m \rangle + \langle l+1, m | \dots | l, m \rangle|^2$$

$$= |\vec{e}_+ a_{l+1}^{m+1} + \vec{e}_- a_{l+1}^{m-1} + \vec{e}_z a_{l+1}^m|^2$$

$$= \frac{1}{2} (a_{l+1}^{m+1})^2 + \frac{1}{2} (a_{l+1}^{m-1})^2 + (a_{l+1}^m)^2 = \frac{l+1}{2l+1}$$

where we have used that $\vec{e}_+ \cdot \vec{e}_+^* = \vec{e}_+ \cdot \vec{e}_- = \frac{1}{2}$, $\vec{e}_+ \cdot \vec{e}_-^* = \vec{e}_+^* \cdot \vec{e}_- = 0$

$$\vec{e}_+ \cdot \vec{e}_z = 0 \quad \& \quad \vec{e}_- \cdot \vec{e}_z = 0$$

$$\Rightarrow M_{nlm, n'(l+1)m'} = \frac{4e^2 \omega_{nn'}^3}{3\hbar c^3} \frac{l+1}{2l+1} \left[\int_0^\infty r^2 dr R_{n'(l+1)}(r) R_{nl}(r) r \right]^2; \quad m'-m = 0, \pm 1$$

And, analogously,

$$M_{nlm, n'(l-1)m'} = \frac{4e^2 \omega_{nn'}^3}{3\hbar c^3} \frac{l}{2l+1} \left[\int_0^\infty r^3 dr R_{n'(l-1)}(r) R_{nl}(r) \right]^2; \quad m'-m = 0, \pm 1$$

~ (Spinless) normal Zeeman effect.

Consider first an atom subject to a weak magnetic field $\vec{B}$. Additionally, suppose that the atom has no spin (the H-atom does have spin). Pieter Zeeman discovered that, depending on the direction & magnitude of $\vec{B}$, each spectral line splits in three



Classically, due to the presence and motion of the electron, it's clear that there must be a magnetic dipole moment $\vec{\mu}$ which can couple to $\vec{B}$, producing a potential term

$$V_z = -\vec{\mu} \cdot \vec{B}$$

If $\vec{\mu}$ is due to the orbiting of e^- around the p^+ in an orbit of radius r and at the rate of f rev/s, the magnitude of $\vec{\mu}$ is

$$\Rightarrow \mu = \frac{1}{2} (\text{area}) (\text{current}) = (\pi r^2) (-ef) = -e f \pi r^2$$

On the other hand, the magnitude of $\vec{L}$ is

$$L = mvr = m(2\pi f r) r = 2m f \pi r^2 = 2m \left(-\frac{\mu}{e}\right)$$

$$\Leftrightarrow \mu = -\frac{e}{2m} L$$

$$\Rightarrow V_z = \frac{e}{2m} \vec{B} \cdot \vec{L}$$

One may be tempted to generalize this result to quantum mechanics

$$\hat{V}_z = \frac{e}{2m} \vec{B} \cdot \hat{L} = \frac{\mu_B}{\hbar} \vec{B} \cdot \hat{L}$$

Bohr magneton
 $\mu_B \equiv \frac{e\hbar}{2m} \sim 5.8 \times 10^{-5} \frac{eV}{T}$

Useful to define the gyromagnetic ratio $\frac{\mu}{L} = g \frac{\mu_B}{\hbar} \rightarrow$ in this case $g = 1$ Landé g-factor

For simplicity, we choose $\vec{B}$ along the z-direction, so that

$$\hat{V}_2 = \frac{\mu_0}{\hbar} B \hat{L}_z$$

The Hamiltonian is then

$$\hat{H} = \hat{H}_0 + \hat{V}_2$$

where

$$\hat{H}_0 \equiv \frac{\hat{p}^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r} \quad \hat{V}_{\text{Coulomb}}(\vec{r})$$

Since the eigenstates of $\hat{H}_0$ ~~satisfy~~ are $|n, l, m\rangle$ and satisfy

$$\hat{L}_z |n, l, m\rangle = m\hbar |n, l, m\rangle$$

then they are also eigenstates of $\hat{H}$.

However, the ^{energy} eigenvalues cannot coincide

$$\hat{H} |n, l, m\rangle = \hat{H}_0 |n, l, m\rangle + \frac{\mu_0 B}{\hbar} \hat{L}_z |n, l, m\rangle$$

$$= \underbrace{\left(E_n^0 + \frac{\mu_0 B}{\hbar} m\hbar \right)}_{E_{n,m}} |n, l, m\rangle$$

$$E_n^0 \equiv -\frac{1}{4\pi^2} \frac{me^4}{8\epsilon_0^2 \hbar^2} \frac{1}{n^2}$$

Thus, the transition frequency shall be

$$\omega_{nn'} \rightarrow \omega_{nm, n'm'} = \frac{E_{n,m} - E_{n',m'}}{\hbar} = \omega_{nn'}^0 + \frac{\mu_0 B}{\hbar} \frac{(m - m')}{\Delta m} = \omega_{nn'}^0 + \frac{\mu_0 B}{\hbar} \Delta m$$

$\Delta m = 0, \pm 1 \rightarrow$ each spectral line splits in three lines with separation $\frac{\mu_0 B}{\hbar}$ & frequencies

$$\omega_{n,n'}^0, \quad \omega_{n,n'}^0 \pm \frac{\mu_0 B}{\hbar}$$

~ Spin & total angular momentum

Observations tell that there must be much more to the Zeeman effect:

- E.g. the first Balmer line ($n=3 \rightarrow n=2$) is actually composed of 2 lines
 - When applying a magnetic field, there appear more than 3 lines!
- Also atoms with $l=0$ could lead to more than 1 line under a magnetic field. In fact, this last issue was called the anomalous Zeeman effect and could not be explained with the simplest atomic theory. There was a missing element: the spin.

By 1922, the existence of an additional contribution $\vec{S}$ to the angular momentum had been demonstrated by Stern & Gerlach, such that the total angular momentum of an atom could be computed as

$$\vec{J} = \vec{L} + \vec{S}$$

A second weird effect was observed when Barnett, Stewart, Beck ^{exp.} verified independently that the gyromagnetic ratio of the electrons ~~was~~ is given ^{is} by

$$\frac{e}{m} = 2 \frac{\mu_B}{\hbar} \equiv g_s \frac{\mu_B}{\hbar} \quad (*)$$

Since in the classical theory

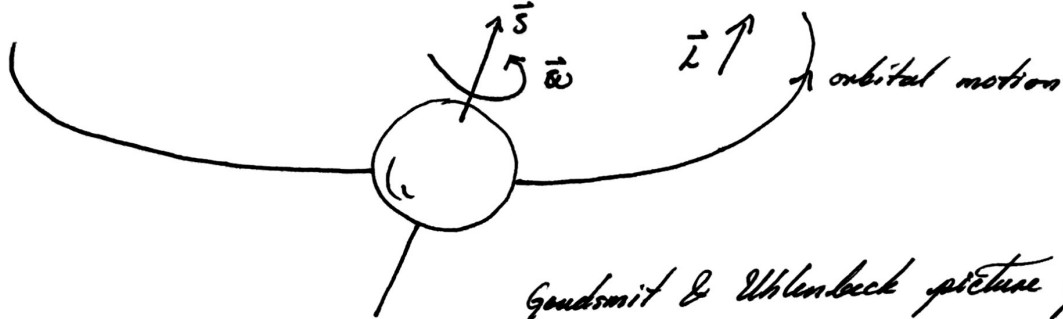
$$\frac{\mu}{L} = g_L \frac{\mu_B}{\hbar} \quad \text{with } g_L = 1,$$

$g_s = 2$ was ~~called the~~ anomalous.

Goudsmit & Uhlenbeck proposed that $\vec{S}$ doesn't correspond to angular momentum of the atom, but an intrinsic angular momentum of the electron itself, whose magnitude ~~produces~~ is independent of the electronic motion. As for $\vec{L}$, G & U suggested that (*) can be interpreted in terms of a new induced magnetic moment

$$\vec{\mu}_s = -\frac{e}{m} \vec{S} = -g_s \frac{\mu_B}{\hbar} \vec{S}, \quad g_s = 2$$

$$(\text{Compare with } \vec{\mu}_L = -g_L \frac{\mu_B}{\hbar} \vec{L}, \quad g_L = 1)$$



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Goudsmit & Uhlenbeck picture of $\vec{L}$ & $\vec{S}$.

Einstein pointed out that this interpretation is wrong from a classical point of view. Particularly, to account for the values of $\vec{S}$ measured from the Stern-Gerlach experiment, one would need a spinning speed greater than c ! However, there's no reason why $\vec{S}$ should be interpreted classically.

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Thus, one expects a potential of the form

$$V = -\vec{\mu}_L \cdot \vec{B} - \vec{\mu}_S \cdot \vec{B} = \frac{\mu_0}{\hbar} (g_L \vec{B} \cdot \vec{L} + g_S \vec{B} \cdot \vec{S}) = \frac{\mu_0}{\hbar} \vec{B} \cdot (\vec{L} + 2\vec{S}) = \frac{\mu_0}{\hbar} \vec{B} \cdot \vec{J}$$

in the presence of a magnetic field.

Clearly, in this scenario, the angular momentum quantum numbers must be four:

$$\hat{L}^2, \hat{L}_z, \hat{S}^2, \hat{S}_z \rightarrow L, m_L, S, m_S$$

One can instead use an equivalent ^{& very useful} set of quantum numbers*

$$\hat{L}^2, \hat{S}^2, \hat{J}^2, \hat{J}_z \rightarrow L, S, J, m_J$$

Notice that since both $\hat{L}_i$ & $\hat{S}_i$ satisfy the same algebra

$$\Rightarrow [\hat{J}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{J}_k$$

$$[\hat{J}^2, \hat{J}_i] = 0$$

$$\Rightarrow \hat{J}^2 |L S J m_J\rangle = J(J+1)\hbar^2 |L S J m_J\rangle$$

$$\hat{J}_z |L S J m_J\rangle = m_J \hbar |L S J m_J\rangle \quad \text{with} \quad -J \leq m_J \leq J$$

In addition, we must find that, as for $\hat{L}_i$,

$$\hat{S}^2 |L S J m_J\rangle = S(S+1)\hbar^2 |L S J m_J\rangle$$

$$\hat{L}^2 |L S J m_J\rangle = L(L+1)\hbar^2 |L S J m_J\rangle$$

* However, we cannot just replace $|L m_L S m_S\rangle$ by $|J m_J\rangle$ in fact $|J m_J\rangle = \sum_{m_L, m_S} C_{L m_L S m_S}^{J m_J} |L m_L S m_S\rangle$ (Clebsch-Gordan)

The values of j are restricted by l & s as follows:

$$|l-s| \leq j \leq l+s$$

(see § 12.4 - 12.6 of de la Peña).

We have seen that for the electron

$$\hat{S} = \frac{1}{2} \hbar \hat{\sigma}$$

$\hat{\sigma}_i$: Pauli matrices

such that

$$[\hat{S}_i, \hat{S}_j] = i \hbar \epsilon_{ijk} \hat{S}_k \Rightarrow [\hat{\sigma}_i, \hat{\sigma}_j] = 2i \epsilon_{ijk} \hat{\sigma}_k$$

$$\Rightarrow \hat{\sigma}_i \hat{\sigma}_j = \delta_{ij} + i \epsilon_{ijk} \hat{\sigma}_k$$

$$\Rightarrow (\hat{\sigma} \cdot \hat{A})(\hat{\sigma} \cdot \hat{B}) = \hat{A} \cdot \hat{B} + i \hat{\sigma} \cdot (\hat{A} \times \hat{B}) \quad \hat{A}, \hat{B} \text{ arbitrary ops.}$$

What is the associated charge to $\hat{S}$? & what shape does $\psi(\mathbf{r})$ take?

Recall that the symmetry operator associated with $\hat{A}$ was defined as

$$\hat{T}(\varphi) = e^{-i\varphi \hat{A}/\hbar}$$

For angular momentum $\hat{L}$ we find

$$\hat{T}(\varphi) = e^{-i\varphi \hat{n} \cdot \hat{L}/\hbar}$$

$\hat{n}$: unitary vector perpendicular to the rotation plane

Thus, since $\hat{S}$ is also angular momentum

$$\hat{T}_S(\varphi) = e^{-i\varphi \hat{n} \cdot \hat{S}/\hbar} = e^{-i\varphi \hat{n} \cdot \hat{\sigma}/2}$$

φ : rotation angle in spin space

which makes sense only as a series. Noting that

Matrix representation $s = \frac{1}{2}$

$$(\hat{n} \cdot \hat{\sigma})^2 = |\hat{n}|^2 \mathbb{I}_2 = 1 \cdot \mathbb{I}_2$$

$$\Rightarrow (\hat{n} \cdot \hat{\sigma})^{2k} = \mathbb{I}_2 \quad \& \quad (\hat{n} \cdot \hat{\sigma})^{2k+1} = \hat{n} \cdot \hat{\sigma} \quad k \in \mathbb{N}$$

$$\begin{aligned} \Rightarrow \hat{T}_S(\varphi) &= \sum_{k=0}^{\infty} \frac{(-i)^k (\hat{n} \cdot \hat{\sigma})^k (\varphi/2)^k}{k!} = \sum_{k=0}^{\infty} \frac{(-1)^k (\hat{n} \cdot \hat{\sigma})^{2k} (\frac{\varphi}{2})^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{-i (-1)^k (\hat{n} \cdot \hat{\sigma})^{2k+1} (\frac{\varphi}{2})^{2k+1}}{(2k+1)!} \\ &= \mathbb{I}_2 \cos \frac{\varphi}{2} - i (\hat{n} \cdot \hat{\sigma}) \sin \frac{\varphi}{2} \end{aligned}$$

$\Rightarrow$ The wavefunction or state associated with this transformation must have two components in matrix representation

$$|\psi\rangle \equiv \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$$

&

$$\hat{J}_z(2\pi) |\psi\rangle = \mathbb{I}_2 \cos \pi |\psi\rangle = -|\psi\rangle$$

$\Rightarrow |\psi\rangle$ is not invariant under a full rotation $\Rightarrow$ spinor

As it turns out, ~~the~~ the spinor $|\psi\rangle$ can be expressed in terms of the usual quantum numbers of angular momentum. To see this, notice first that, for $|\psi\rangle \equiv \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$\hat{S}_z |\psi\rangle \equiv \hat{S}_z \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = +\frac{1}{2} \hbar \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \hbar |\psi\rangle$$

whereas for $|\psi\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$\hat{S}_z |\psi\rangle \equiv \frac{1}{2} \hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{2} \hbar |\psi\rangle$$

i.e. in terms of ~~the theory of moment~~ ang. mom. quantum numbers corresponds to

$$m_s = \pm \frac{1}{2} \quad \Rightarrow \quad \hat{S}_z |\psi\rangle = m_s \hbar |\psi\rangle$$

Also $\hat{S}^2 |\psi\rangle = \frac{1}{4} \hbar^2 (\hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2) |\psi\rangle = \frac{3}{4} \hbar^2 |\psi\rangle = \frac{1}{2} (\frac{1}{2} + 1) \hbar^2 |\psi\rangle$
 We can thus introduce a new quantum number s , called spin, such that (by extrapolation of $\hat{L}$)

$$\hat{S}^2 |\psi\rangle = s(s+1) \hbar^2 |\psi\rangle$$

$$\hat{S}_z |\psi\rangle = m_s \hbar |\psi\rangle$$

$$-s \leq m_s \leq s$$

& $|\psi\rangle \equiv |s, m_s\rangle$ is the spinor

For the electron $s = \frac{1}{2}$ & thus $m_s = \pm \frac{1}{2}$, as observed.

All particles with half-integer s are called fermions!

" " " integral s " " bosons!

→ (82)

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→ & fine structure of Hydrogen spectrum

~ ~~Spin-orbit interaction~~: $\vec{L} \cdot \vec{S}$ coupling in ~~Hydrogen spectrum~~

Due to the existence of an "external" (to the electron) electric field, the electron moving at speed $v \ll c$ "feels" a magnetic field

$$\vec{B}_{\text{eff}} = -\frac{\vec{v}}{c^2} \times \vec{E} = -\frac{1}{mc^2} \vec{p} \times \vec{E} \quad \text{with} \quad \vec{E} = -\frac{1}{e} \vec{\nabla} V(r) = -\frac{1}{e} \vec{e}_r \frac{dV}{dr} = -\frac{1}{e} \frac{\vec{r}}{r} \frac{dV}{dr}$$

$$= \frac{1}{mc^2 r} \frac{dV}{dr} \frac{\vec{p} \times \vec{r}}{-1} = -\frac{1}{mc^2 r} \frac{dV}{dr} \vec{L} = -\frac{1}{4\pi\epsilon_0} \frac{e}{r^3} \vec{r}, \quad V(r) = \frac{-e^2}{4\pi\epsilon_0 r}$$

$$\Rightarrow \vec{B}_{\text{eff}} = +\frac{1}{mc^2} \frac{e}{4\pi\epsilon_0 r^3} \vec{p} \times \vec{r} = -\frac{e \vec{L}}{4\pi\epsilon_0 mc^2 r^3}$$

$$\Rightarrow V_{LS} = -\vec{\mu}_s \cdot \vec{B}_{\text{eff}} = -\left(-\frac{e}{m} \vec{S}\right) \cdot \left(-\frac{e}{4\pi\epsilon_0 mc^2 r^3} \vec{L}\right) = -\left(-\frac{e}{m} \vec{S}\right) \cdot \left(-\frac{1}{mc^2 r} \frac{dV}{dr} \vec{L}\right)$$

$$= -\frac{e^2}{4\pi\epsilon_0 m^2 c^2 r^3} \vec{L} \cdot \vec{S}$$

Thomas factor

$$V_{LS} = -\frac{1}{2} \vec{\mu}_s \cdot \vec{B}_{\text{eff}} = \frac{-e^2}{8\pi\epsilon_0 m^2 c^2 r^3} \vec{L} \cdot \vec{S}$$

$$= \frac{1}{2} (\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$$

Using $\vec{J} = \vec{L} + \vec{S}$, we find

$$V_{LS} = \frac{e^2}{8\pi\epsilon_0 m^2 c^2 r^3} (\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$$

This equation is almost right. Thomas showed that this result could be obtained from special relativity (see e.g. §15.4 de la Peña)

except that in that scenario there's an extra factor $\frac{1}{2}$ which was called Thomas half or Thomas factor. Thomas gave also an explanation in terms of the (relativistic) precession of the electron spin

$$\Rightarrow V_{LS} = \frac{-e^2}{8\pi\epsilon_0 m^2 c^2 r^3} \vec{L} \cdot \vec{S}$$

It's useful first to notice that using $\vec{J} = \vec{L} + \vec{S}$

$$\vec{L} \cdot \vec{S} = \frac{1}{2} (\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$$

In the hydrogen system, we must promote $V_{ls} \rightarrow \hat{V}_{ls}$

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$$\Rightarrow \begin{aligned} r &\rightarrow \hat{r} \\ \vec{L} &\rightarrow \hat{\vec{L}} \\ \vec{S} &\rightarrow \hat{\vec{S}} \end{aligned} \quad \text{Note that} \quad * [\hat{\vec{L}} \cdot \hat{\vec{S}}, \hat{J}_i] = 0 \quad \text{BUT} \quad [\hat{\vec{L}} \cdot \hat{\vec{S}}, \hat{L}_i] \neq 0$$

$\rightarrow$ choice of $|l s j m_j\rangle$ nice!

In order to compute the corrections to the energies & thus to the spectra, we can't simply use the eigenvalues eg. since

Perf. $E = E_n^0 + \langle n l s j m_j | V_{ls} | n l s j m_j \rangle$ if $\langle V_{ls} \rangle \ll \langle H_0 \rangle$

$$\frac{1}{r^3} |n l s j m_j\rangle \neq \frac{1}{r^3} |n l s j m_j\rangle$$

doesn't yield an eigenvalue of $|n l s j m_j\rangle$ (these states are eigenstates of energy, not position). But we can compute the expectation value

$$\langle n l s j m_j | \hat{V}_{ls} | n l s j m_j \rangle$$

$$= - \frac{e^2}{8\pi\epsilon_0 m^2 c^2} \langle n l | \frac{1}{r^3} | n l \rangle \langle l s j m_j | \hat{\vec{L}} \cdot \hat{\vec{S}} | l s j m_j \rangle$$

One can show that*

$$\langle n l | \frac{1}{r^3} | n l \rangle = \int_0^\infty dr \frac{1}{r^3} (R_{nl}(r))^2 = \frac{1}{a_0^3} \frac{1}{n^3 l(l+1)(l+\frac{1}{2})}$$

whereas

$$\begin{aligned} \langle l s j m_j | \hat{\vec{L}} \cdot \hat{\vec{S}} | l s j m_j \rangle &= \frac{1}{2} \langle l s j m_j | \hat{J}^2 - \hat{L}^2 - \hat{S}^2 | l s j m_j \rangle \\ &= \frac{1}{2} \langle l s j m_j | (j(j+1) - l(l+1) - s(s+1)) \hbar^2 | l s j m_j \rangle \\ &= \frac{1}{2} [j(j+1) - l(l+1) - s(s+1)] \hbar^2 \end{aligned}$$

$$\Rightarrow E_{n l s} = E_n - \frac{e^2 \hbar^2}{16\pi\epsilon_0 m^2 c^2 a_0^3} \frac{j(j+1) - l(l+1) - s(s+1)}{n^3 l(l+1)(l+\frac{1}{2})}$$

$$\text{for } \hat{H} = \hat{H}_0 + \hat{V}_{ls}$$

* For $l=0$, this is not a valid result. In fact, considering relativistic corrections one can show that $\langle n, l=0, s j m_j | \hat{V}_{ls} | n, l=0, s j m_j \rangle = 0$.

Notice that, with $a_0 = \frac{\epsilon_0 \hbar^2}{m e^2 \pi} = \frac{\hbar}{\alpha_{em} m c}$ $E_n = -\frac{m e^4}{8 \epsilon_0^2 \hbar^2 n^2} = -\frac{m c^2 \alpha_{em}^2}{2 n^2}$

$$-\frac{e^2 \hbar^2}{8 \pi \epsilon_0 m^2 c^2 a_0^3} \frac{1}{n^2} = -\frac{m e^4}{8 \epsilon_0^2 \hbar^2 n^2} \left(\frac{e^2}{2 \epsilon_0 \hbar c} \right)^2$$

$$= + E_n \left(\frac{e^2}{4 \pi \epsilon_0 \hbar c} \right)^2$$

$$\equiv \alpha_{em} \sim \frac{1}{137} \quad \checkmark \quad \begin{array}{l} \text{measured value} \\ \text{fine structure constant} \end{array}$$

$\Rightarrow$

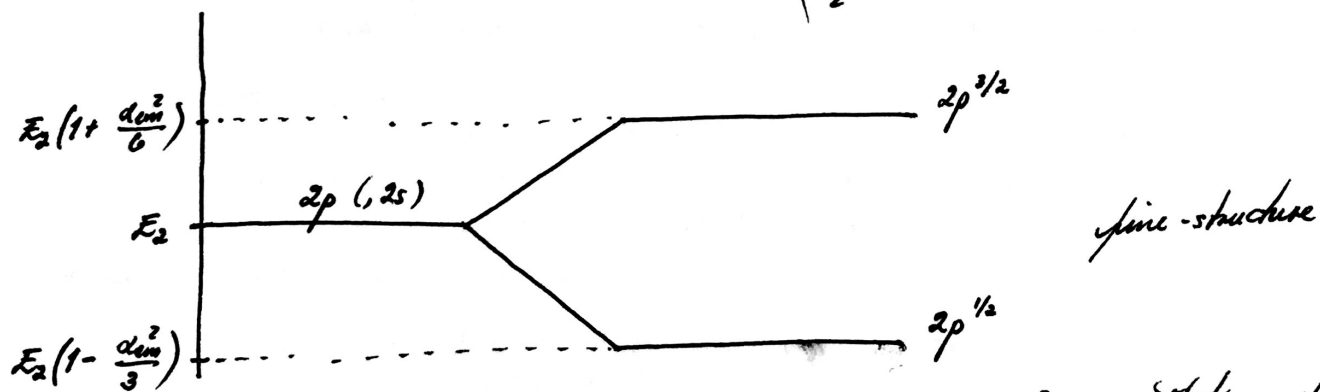
$$E_{njls} = E_n + E_n \frac{\alpha_{em}^2}{2n} \frac{j(j+1) - l(l+1) - s(s+1)}{l(l+1)(l+\frac{1}{2})} \equiv E_n + \Delta E_{njls}^{LS}$$

As an example, consider $n=2$ & $l=1$ (for the first Balmer line)

$$l=1, s=\frac{1}{2} \Rightarrow |l-s| \leq j \leq l+s \Rightarrow j = \begin{cases} \frac{3}{2} \\ \frac{1}{2} \end{cases}$$

$$j = \frac{3}{2} : \Delta E_{njls}^{LS} = E_2 \frac{\alpha_{em}^2}{2} \frac{\frac{3}{2}(\frac{5}{2}) - 2 - \frac{1}{2}(\frac{3}{2})}{1 \cdot 2 \cdot \frac{3}{2}} = E_2 \frac{\alpha_{em}^2}{2} \frac{1}{3}$$

$$j = \frac{1}{2} : \Delta E_{njls}^{LS} = E_2 \frac{\alpha_{em}^2}{2} \frac{\frac{1}{2}(\frac{3}{2}) - 2 - \frac{1}{2}(\frac{1}{2})}{2 \cdot \frac{3}{2}} = -E_2 \frac{\alpha_{em}^2}{2} \frac{2}{3}$$



Note that this fine-structure is prop. to α_{em}^2 . That's why this constant receives that name.

(OBS. Sometimes, it's convenient to use units such that $\hbar = c = 1 \Rightarrow \epsilon_0 = 1$)
In these units
 $\alpha_{em} = \frac{e^2}{4\pi}$ Typical in HEP